Various water recipes

From German brewing and more

These water recipes are guidelines that brewers, who build their own water, can follow. None of these recipes have been optimized for a given style but the residual alkalinity is chosen such that it should yield a satisfactory mash pH when used for the listed styles. Since the acidity of the grist can vary significantly even for a given style additional adjustments might be necessary. All water recipes include a copy of my water calculation spreadsheet which can be used to make such adjustments. In that spreadsheet you may also enter a starting water profile if that is known.

Some of the recipes used magnesium chloride. I got this salt from a fellow brewer who also keeps fish and used it to build water for the aquarium. If you don't have access to $MgCl_2$, don't worry and just omit it or make up for it with some more Epsom salt.

Some of these recipes give options for the use of dissolved chalk. Check out Building brewing water with dissolved chalk if you want to go that route.

Contents

very soft water

This is a very soft water option for a Munich Helles or a Pilsner. I have used this for one batch and despite the low calcium level did not have problems with the getting the beer to clear. Paulaner, for example, uses a deep well from which they get very soft (~ 2 dH / 30 ppm CaCO3 Hardness) brewing water. I don't know if they are adding additional minerals when they brew their lighter beers with that water but it is very likely that they don't.

I use this water with a 97% Pilsner malt and 3% acidulated malt (Sauermalz) grist and get a mash pH of about 5.3-5.4.

water profile

ppm Ca	ppm Mg	ppm Na	ppm SO4	ppm Cl	ppm HCO3	alkalinity as ppm CaCO3	residual alkalinity as ppm CaCO3
22	8	0	31	39	0	0	-20

salts

Gypsym	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved	
CaSO4 ·2H2O	MgSO4 ·7H2O	NaCl	CaCl2 ·2H2O	MgCl2 ·6H2O	NaHCO3	CaCO3	CaCO3 + CO2	
0	80	0	80	0	0	0	0	ppm
0	0.08	0	0.08	0	0	0	0	g/l
0	0.3	0	0.3	0	0	0	0	g/gal

works for:

- Munich Helles
- Weissbier Hell (light Bavarian Wheat)
- Pilsner
- Koelsch style ales

Brewing school Weihenstephan water

It is not often that I come across actual brewing water analysis data in the literature but the Markus Hermann's dissertation (http://deposit.d-nb.de/cgi-bin/dokserv?idn=978186087&dok_var=d1&dok_ext=pdf&filename=978186087.pdf) gave an insight into the water that is used for brewing studies at the Weihenstephan brewing school in Freising, Germany.

While it makes a great water for Weissbier, in fact the dissertation was about brewing Weissbier, it is not necessarily the water that the Weihenstephan brewery uses.

with unsdissolved chalk:

water profile

ppm Ca	ppm Mg	ppm Na	ppm SO4	ppm Cl	ppm HCO3	alkalinity as ppm CaCO3	residual alkalinity as ppm CaCO3
15	10	0	17	17	46	37	21

salts

Gypsym	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved	
CaSO4 ·2H2O	MgSO4 ·7H2O	NaCl	CaCl2 ·2H2O	MgCl2 ·6H2O	NaHCO3	CaCO3	CaCO3 + CO2	
0	43	0	0	50	0	75	0	ppm
0	0.04	0	0	0.05	0	0.08	0	g/l
0	0.16	0	0	0.19	0	0.28	0	g/gal

with chalk dissolved with CO2:

water profile

ppm Ca	ppm Mg	ppm Na	ppm SO4	ppm Cl	ppm HCO3	alkalinity as ppm CaCO3	residual alkalinity as ppm CaCO3
16	10	0	17	17	49	40	23

salts

Gypsym	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved	
CaSO4 ·2H2O	MgSO4 ·7H2O	NaCl	CaCl2 ·2H2O	MgCl2 ·6H2O	NaHCO3	CaCO3	CaCO3 + CO2	
0	43	0	0	50	0	0	40	ppm
0	0.04	0	0	0.05	0	0	0.04	g/l
0	0.16	0	0	0.19	0	0	0.15	g/gal

water_weihenstephan_dissolved_chalk.xls (http://braukaiser.com/documents/water_profiles/water_weihenstephan_dissolved_chalk.xls)

works well for:

- Munich Helles (with the use of 2-3% acid malt in the grist)
- Weissbier (light and amber varieties)

Pilsner water

This water I have used for German Pilsner and American Pilsner with success. It has no alkalinity and a calcium level of ~60 ppm. This water, Pilsner malt and 2% acid malt in the grist should yield a mash pH round 5.4.

water profile

ppm Ca	ppm Mg	ppm Na	ppm SO4	ppm Cl	ppm HCO3	alkalinity as ppm CaCO3	residual alkalinity as ppm CaCO3
59	8	0	89	63	0	0	-47

salts

Gypsym	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved	
CaSO4 ·2H2O	MgSO4 ·7H2O	NaCl	CaCl2 ·2H2O	MgCl2 ·6H2O	NaHCO3	CaCO3	CaCO3 + CO2	
100	85	0	130	0	0	0	0	ppm
0.1	0.09	0	0.13	0	0	0	0	g/l
0.38	0.32	0	0.49	0	0	0	0	g/gal

works well for:

- German Pilsner
- Classic American Pilsner
- Weissbier (lightly colored types)

Moderately alkaline water

This is the water I tend to used for dark German beers, in paticular for my Schwarzbier. There is a undissolved and dissolved chalk option.

with unsdissolved chalk:

water profile

ppm Ca	ppm Mg	ppm Na	ppm SO4	ppm Cl	ppm HCO3	alkalinity as ppm CaCO3	residual alkalinity as ppm CaCO3
80	10	26	16	33	165	136	73

salts

Gypsym	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved	
CaSO4 ·2H2O	MgSO4 ·7H2O	NaCl	CaCl2 ·2H2O	MgCl2 ·6H2O	NaHCO3	CaCO3	CaCO3 + CO2	
0	40	25	0	50	60	200	0	ppm
0	0.04	0.03	0	0.05	0.06	0.2	0	g/l
0	0.15	0.09	0	0.19	0.23	0.76	0	g/gal

with chalk dissolved with CO2:

water profile

ppm Ca	ppm Mg	ppm Na	ppm SO4	ppm Cl	ppm HCO3	alkalinity as ppm CaCO3	residual alkalinity as ppm CaCO3
40	10	21	16	33	151	124	89

salts

Gypsym	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved	
CaSO4 ·2H2O	MgSO4 ·7H2O	NaCl	CaCl2 ·2H2O	MgCl2 ·6H2O	NaHCO3	CaCO3	CaCO3 + CO2	
0	40	25	0	50	40	0	100	ppm
0	0.04	0.03	0	0.05	0.04	0	0.1	g/l
0	0.15	0.09	0	0.19	0.15	0	0.38	g/gal

works well for:

- Schwarzbier
- Doppelbock
- Weissbier (very dark varieties)

Munich Water

To create this water I went to the web page of the water department of the city of Munich and read the current water report (http://www.swm.de/dokumente/swm/pdf/wasser/trinkwasserwerte.pdf). I was able to come close but could have come closer with the use of Magnesium carbonate. But in the end that won't matter much. This water capures the essence of the Munich water: high temporary hardness and low permanent harness. In English, there are much more bicarbonate ions in this water than chloride or sulfate. I use this water for dark Bavarian beers like Bock and Doppelbock.

While this is mimics the city water of Munich I don't know how many breweries actually brew with this water. Since it works well for dark beers, I would assume that a Munich brewer, who doesn't have his own water supply, would be using it without any treatment to brew these beers.

This recipe uses magnesium chloride. If you don't have that salt just leave it out.

with unsdissolved chalk:

water profile

ppm Ca	ppm Mg	ppm Na	ppm SO4	ppm Cl	ppm HCO3	alkalinity as ppm CaCO3	residual alkalinity as ppm CaCO3
160	8	3	17	10	251	206	87

salts

Gypsym	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved	
CaSO4 ·2H2O	MgSO4 ·7H2O	NaCl	CaCl2 ·2H2O	MgCl2 ·6H2O	NaHCO3	CaCO3	CaCO3 + CO2	
0	43	0	0	30	10	400	0	ppm
0	0.04	0	0	0.03	0.01	0.4	0	g/l
0	0.16	0	0	0.11	0.04	1.51	0	g/gal

with chalk dissolved with CO2:

water profile

ppm Ca	ppm Mg	ppm Na	ppm SO4	ppm Cl	ppm HCO3	alkalinity as ppm CaCO3	residual alkalinity as ppm CaCO3
80	8	3	17	10	251	206	144

salts

Gypsym	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved	
CaSO4 ·2H2O	MgSO4 ·7H2O	NaCl	CaCl2 ·2H2O	MgCl2 ·6H2O	NaHCO3	CaCO3	CaCO3 + CO2	
0	43	0	0	30	10	0	200	ppm
0	0.04	0	0	0.03	0.01	0	0.2	g/l
0	0.16	0	0	0.11	0.04	0	0.76	g/gal

works well for:

- Traditional Bock
- Doppelbock
- Weissbier Dark
- Schwarzbier